

MAHALAKSHMI J

Software Developer | AI/ML Engineer | Computer Vision Developer

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PROFESSIONAL SUMMARY

Software Developer and AI/ML Engineer with hands-on experience in developing intelligent applications using Python, Deep Learning, Computer Vision, and NLP. Experienced in building software solutions across healthcare, accessibility, and automation domains. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, and Software Engineering, with a focus on developing scalable and efficient applications.

EDUCATION

MANGAYARKARASI COLLEGE OF ENGINEERING

2023 – 2027

Bachelor of Engineering in Computer Science and Engineering (AI & ML).

TECHNICAL SKILLS

Programming Languages:	Python, Java, C
AI/ML:	Machine Learning, Deep Learning, Generative AI, Natural Language Processing (NLP), Computer Vision, Data Analytics
Frameworks & Libraries:	TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Flask
Core CS:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), Software Engineering, Database Management Systems (DBMS), Operating Systems
Cloud Platforms:	Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
Web Development:	HTML, CSS, JavaScript
Databases:	PostgreSQL, SQLite
Tools & Platforms:	Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Odoo CRM

PROJECTS

- Brain Tumor Classification using CNN** – Developed a CNN-based application for MRI image classification and brain tumor detection using Python and Deep Learning techniques.
- Automated Attendance System** – Built a facial recognition-based attendance management application using Computer Vision and Python.
- Traffic Sign Recognizer** – Designed and developed a real-time traffic sign detection and classification system for intelligent driving assistance.
- Eye/Hand Controlled Mouse** – Developed an accessibility-focused virtual mouse application using gesture and gaze tracking techniques.
- Air Writing Character Recognizer** – Created a Computer Vision-based application to recognize characters written in air through gesture tracking.
- AI-Based Pest Detection System** – Designed an AI-powered crop pest detection application for precision agriculture and monitoring.

PROFESSIONAL EXPERIENCE

INTERNSHIPS:

- Python Development Intern** – Tarcin Robotic
- AI Intern** – Microsoft Edunet Foundation, Novitech R&D Pvt Ltd
- Machine Learning Intern** – Novitech R&D Pvt Ltd
- Full Stack Development Intern** – Novitech R&D Pvt Ltd
- Data Analytics Intern** – Novitech R&D Pvt Ltd, Adinn Digital

RESEARCH PAPERS

- VR Braille Education Hub
- Digital Banking for the Visually Impaired
- AI-Powered Integrated Smart Traffic Management and Safety System
- AI-Driven Virtual Tutor and Teacher Assistant
- AI-Powered Global Epidemic Prediction and Early Warning System

CERTIFICATIONS

- AI, ML & Full Stack Development** – Novitech R&D Pvt Ltd
- AI for Management** – NPTEL
- AI & Generative AI Basics** – Microsoft
- Generative AI & Prompt Engineering** – IBM
- Artificial Intelligence** – Samsung Innovation Campus

LEADERSHIP & ACHIEVEMENTS

- Vice President** – Data Science Club, Mangayarkarasi College of Engineering
- Led coordination of technical events, workshops, and collaborative student initiatives.
- Organized technical seminars, workshops, and AI-related events for students.
- Participated in AI/ML project development and research activities.
- Collaborated with peers on innovative technology solutions.

LANGUAGES KNOWN

- Tamil – Native
- English – Fluent
- Hindi – Basic